



NUCLEAR SUPPLIES MANUFACTURER

The Indian nuclear supplies manufacturer plays a pivotal role in ensuring the success of the nation's nuclear program. Responsible to produce nuclear fuel and reactor components, they operate multiple specialized plants with distinct production requirements. In order to boost efficiency and be on par with international standards, they aimed to modernize their laboratory and production management systems.

Agaram Technologies proposed a tailored solution leveraging the upgraded **Qualis LIMS** platform built on the React JS framework to meet the manufacturer's requirements. With enhanced scalability, security, and compliance capabilities, Qualis LIMS has set a benchmark for operational excellence in the nuclear industry and positioned them to meet the demands of a rapidly evolving technological landscape.



WHAT'S UP WITH QUALITY CONTROL LABORATORIES

An integral Indian nuclear supplies manufacturer is responsible for the production of nuclear fuel assemblies and reactor core components. Its operations are vital for supporting India's power generation capabilities and nuclear technology advancements.

Their production infrastructure is divided into multiple plants, each responsible for manufacturing specific products. These include fuel assemblies, zirconium alloys, and other critical reactor materials. Each plant functions independently with separate data systems tailored to its unique production processes.

The production and laboratory operations at the nuclear supplies manufacturer are heavily regulated, requiring strict adherence to quality standards, traceability, and data security. To achieve higher efficiency and align operations with global standards, they sought to upgrade its laboratory and production management systems. While they had employed a Flex-based Qualis LIMS for its laboratory operations, the production areas still relied on legacy systems with limited connectivity and integration, presenting challenges for seamless operations.

CHALLENGES

Lack of Integration Between Systems: Production data and laboratory test results were managed in silos, making it difficult to correlate manufacturing data with quality assurance reports. No automated mechanism existed for data exchange between the production systems and the laboratory LIMS.

Inadequate User Access Controls: User access was not segregated by plant, leading to security vulnerabilities. A user in one plant could potentially access data from another plant, breaching operational confidentiality.

Limited Operational Efficiency: Manual data handling and a lack of streamlined workflows resulted in delays and increased the risk of errors in both production and laboratory operations.

GET THE PICTURE

In the production areas, legacy systems were used to manage user records, operational data, and production workflows. However, these systems operated in isolation and lacked integration with laboratory systems. Meanwhile, the laboratories relied on legacy Qualis LIMS to

handle sample testing workflows, including quality checks for raw materials and finished products. The lack of communication between production and lab systems created silos, limiting efficiency and data traceability across operations.

To address these operational gaps and to future-proof their infrastructure, the nuclear supplies manufacturer defined their requirements as given below:

Integrated Data Management: Unify production and laboratory systems for seamless data exchange and centralized management.

Plant-Specific User Access: Implement user access controls that restrict users to their respective plants, ensuring data security and compliance.

Enhanced Workflow Automation: Automate data flows between production and laboratories, minimizing manual intervention and improving accuracy.

Scalable Infrastructure: Upgrade to a system capable of accommodating future expansions and evolving operational needs.

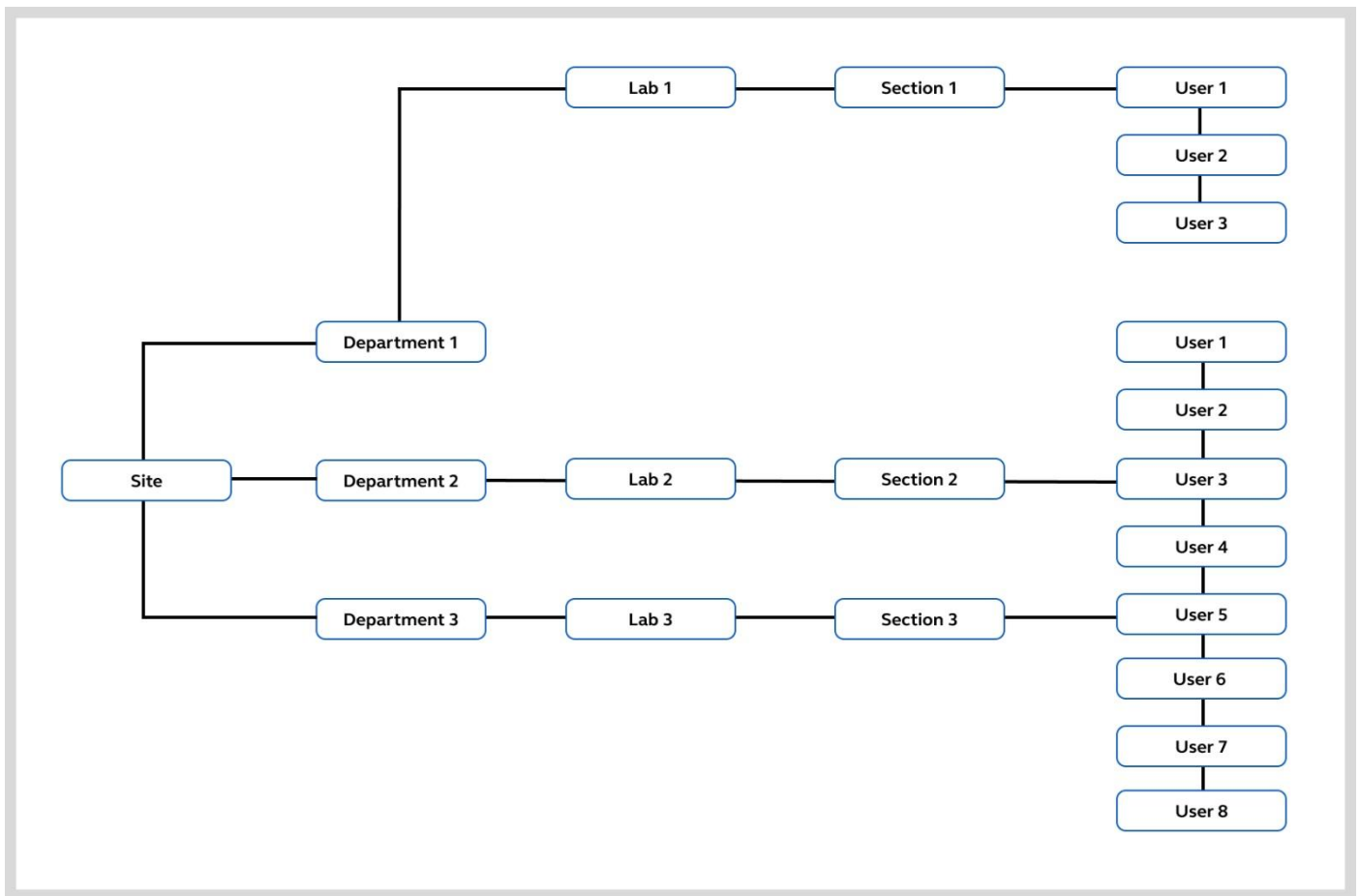
Quality Standards & Regulatory Compliance: Ensure the new system supports robust audit trails, traceability, and regulatory compliance.

ELEVATED OPERATIONAL PERFORMANCE

To address the need for a unified, scalable, and secure system that could integrate production and laboratory operations, Agaram Technologies proposed a tailored solution leveraging the upgraded [Qualis LIMS](#) platform, built on the robust React JS framework. The implementation process for the upgraded Qualis LIMS at the nuclear supplies manufacturer was a comprehensive, multi-phase initiative designed to ensure minimal disruption to ongoing operations while addressing all identified challenges and requirements.

The project involved in-depth requirement gathering through stakeholder workshops, followed by the design of a dual-architecture system to enable plant-specific access and real-time data exchange between production and laboratory environments. Historical data from legacy Flex-based systems was carefully mapped, validated, and migrated to ensure completeness and accuracy. The final phase included rigorous functional, performance, and security testing to validate the system's readiness.

This end-to-end implementation resulted in a future-ready, integrated platform that enhances operational efficiency, data integrity, and regulatory compliance.



Organizational Workflow

ADVANTAGES OF UPGRADING QUALIS LIMS

The transition from the Flex-based architecture to the React JS-based version of Qualis LIMS introduces a host of enhancements that significantly improve system performance, scalability, and user experience. These benefits align with the evolving operational needs of the nuclear supplies manufacturer, positioning the organization for greater efficiency and innovation in laboratory information management.

- **Enhanced Performance and Scalability:** The React JS-based system adopts a microservices architecture, allowing the nuclear supplies manufacturer to scale operations seamlessly. The hybrid database model, integrating both SQL and NoSQL databases, ensures faster data retrieval and storage, reducing latency even under heavy workloads. This scalability enables them to accommodate expanding datasets and integrate additional facilities without disrupting existing workflows.
- **Modern User Interface:** The React JS framework provides a dynamic and responsive user interface. This upgrade significantly enhances usability by delivering an intuitive experience across devices, including desktops, tablets, and mobile platforms. Users benefit from faster page loading times and a consistent interface that improves productivity.
- **Robust Integration Capabilities:** The upgraded system supports dynamic API creation, simplifying the integration of laboratory instruments and external systems. This ensures seamless interoperability between production and laboratory data, aligning with nuclear supplies manufacturer's objective of unified operations. Automated data exchange reduces manual errors, enhancing accuracy and efficiency.
- **Advanced Data Management:** The new architecture enables efficient handling of structured and unstructured data through a hybrid database design. Features such as advanced query tools and real-time data visualization provide actionable insights, empowering decision-makers with a comprehensive view of laboratory processes.

- **Improved Security:** Upgrading to React JS enhances system security through features like role-based access controls, multi-factor authentication, and advanced data encryption. This ensures compliance with stringent regulatory standards and safeguards sensitive data, addressing nuclear supplies manufacturer's requirement for plant-specific user access and data integrity.
- **Future-Ready Architecture:** The new upgrade positions the nuclear supplies manufacturer for long-term technological advancements. Its modular design supports continuous updates and extensions, allowing them to adopt emerging technologies such as artificial intelligence and machine learning for predictive analytics and process optimization.
- **Cost Efficiency:** By streamlining workflows and automating processes, the upgraded LIMS reduces operational overheads. The modular architecture also allows the nuclear supplies manufacturer to implement features incrementally, optimizing investment in line with operational priorities.
- **Regulatory Compliance:** The React JS version adheres to international standards like ISO 17025 and other industry-specific statutory regulatory requirements, ensuring robust audit trails and traceability. This compliance facilitates smooth regulatory audits and reinforces the nuclear supplies manufacturer's commitment to maintaining high-quality standards.

By adopting the React JS-based Qualis LIMS, the nuclear supplies manufacturer stands to significantly enhance their laboratory and production management capabilities. The upgrade not only addresses existing challenges but also lays a strong foundation for future growth, ensuring they remain at the forefront of innovation in nuclear fuel production.

KEY BENEFITS

90% ↑

Operational
Efficiency

80% ↑

Data
Integrity

85% ↑

Integration
Capability

79% ↑

Workflow
Automation

- **Enhanced Security:** Plant-specific access controls have eliminated unauthorized data access, ensuring operational confidentiality.
- **Integrated Operations:** Unified workflows between production and laboratory areas have streamlined operations, reducing delays and errors.
- **Improved Efficiency:** Automation of data exchange and workflow processes has significantly enhanced productivity and reduced manual workload.
- **Data Integrity and Compliance:** Robust audit trails and traceability features ensure compliance with regulatory standards and improve audit readiness.
- **Scalability and Future-Readiness:** The modernized platform supports the nuclear supplies manufacturer's future growth, enabling seamless integration of additional plants and advanced technologies.
- **Real-Time Insights:** The system provides real-time access to critical data, empowering decision-makers to act swiftly and confidently.

By addressing critical challenges such as data silos, security vulnerabilities, and inefficiencies, Agaram Technologies' implementation of the upgraded Qualis LIMS in React JS at the nuclear supplies manufacturer marked a significant milestone in modernizing its operations.

To see [Qualis LIMS](#) action, request a demo and get a detailed overview [here](#).

Compliance-Ready | Digitally Traceable | Scalable | Secure

Automate and streamline your lab operations today!



Global HQ

Raheja Towers, 510 Alpha Wing,
Anna Salai, Chennai - 600 002,
Tamil Nadu, India.

T: +91 44 4208 2005

T: +91 44 42189406

E: info@agaramtech.com

South Korea

941 Hogue-Geumjeong SK V1
Center, 42, LS-ro, Dongan-gu,
Anyang-si, Gyeonggi-do 14118,
South Korea.

T: +82 31 429 8958

E: info@agaramtech.com

For product enquiries & questions:
sales@agaramtech.com

Official Website:
www.agaramtech.com

Logilab ELN Website:
www.logilabeln.com